



Equb Management Platform

Digital infrastructure for Ethiopia's rotating savings groups

113+ Spots / Cycle

Weekly Draws

SMS Alerts

Role-Based Access

What is Equb?

The Traditional Concept

Equb (አቁብ) is a centuries-old Ethiopian rotating savings and credit group (ROSCA).

A fixed group of members each contribute an agreed amount every week.

Each cycle, one member wins the entire pot through a draw.

Every member eventually receives the full pot — guaranteed.

It is built on trust, community, and mutual financial support.

Why a Digital Platform?

✗ Manual tracking → errors and disputes

✗ No audit trail for payments and draws

✗ Hard to scale beyond ~20 members

✗ SMS reminders impossible at scale

✓ Automated payment status tracking

✓ Transparent weekly draw records

✓ 113 member spots per cycle

✓ Africa's Talking SMS integration

✓ Role-based admin / cashier access

Key Features

Member Management

- Track 113+ member spots + 5 association spots
- Full & half-share spot splitting
- Member status: active / left / stopped paying

Cycles & Weekly Draws

- Multi-cycle support with per-cycle settings
- Automated draw scheduling every week
- Status auto-transitions: pending → late → missed

Payment Tracking

- Weekly payment entry per member
- Batch payment recording by cashiers
- Payment history with late/missed flags

Disbursements

- Cheque-based payouts with 3 guarantors required
- Voucher deductions tracked per member
- End-of-cycle distribution cheques

Pot Sales & Transfers

- Group week sales, member-to-member transfers
- Association spot sales with profit routing
- Net pot = gross minus association deduction

SMS Notifications

- Africa's Talking API integration
- 6 templates: reminder, winner, missed, on-hold...
- Auto-reminders 48 hours before each draw

Reports & Analytics

Financial Summary	Total collected vs. expected per cycle; pot size breakdown; association fund balance
Payment Status Report	Week-by-week member payment grid: paid / late / missed per member
Member Status Report	Active, exited, and stopped-paying members with join/exit dates
Draw History	All past draws: winner, week number, pot amount, guarantors
Disbursement Ledger	Cheque records, voucher deductions, guarantor list per payout
Collection Trend Chart	Actual ETB collected vs. expected — color-coded weekly bar chart
SMS Log	Sent notification history per member: type, timestamp, status
Association Expense Log	Vendor payments and expense categories for the association fund

Technology Stack

Frontend	Backend	Database	Services
Jinja2 HTML templates	FastAPI (Python)	PostgreSQL (production)	Africa's Talking SMS API
Tailwind CSS	Uvicorn ASGI server	SQLite (local dev)	Railway.app hosting
Chart.js visualisations	APScheduler (cron jobs)	Railway-hosted Postgres	PBKDF2 password hashing
Vanilla JavaScript	SQLAlchemy ORM	Alembic migrations	CSP / HTTPS / CSRF

Security & Access Control

Role-Based Access

Superadmin

Full access: settings, users, all data, reports

Admin

Cycle management, draws, disbursements, reports

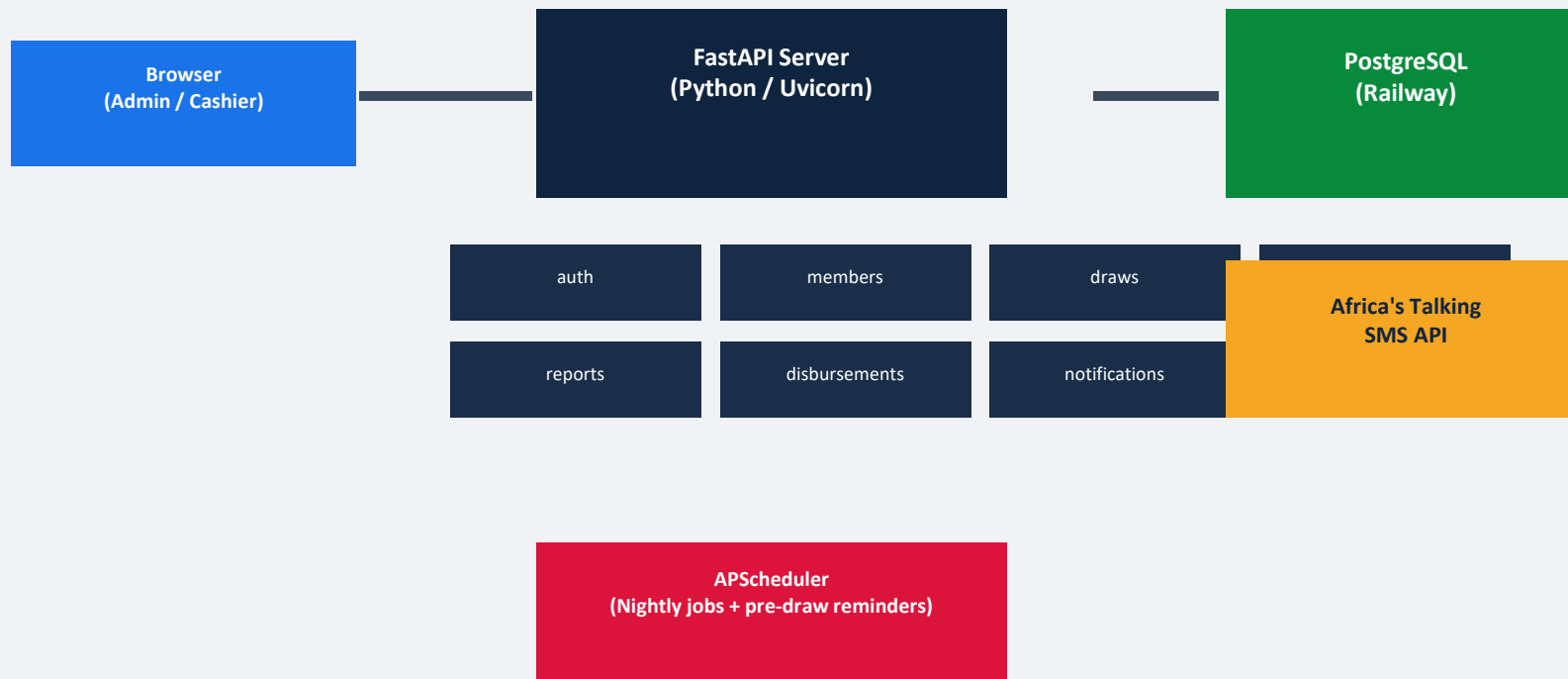
Cashier

Payment entry and member viewing only

Security Measures

- ✓ PBKDF2 password hashing (no plain-text storage)
- ✓ Rate limiting: 5 failed logins → 5-minute lockout
- ✓ CSRF protection on all state-changing requests
- ✓ Content Security Policy (CSP) headers enforced
- ✓ HTTPS enforcement in production
- ✓ Session middleware with secure cookie flags
- ✓ Granular per-role permission matrix
- ✓ Audit trail for payments, draws, and disbursements

System Architecture



Hosted on [Railway.app](#) | [Auto-deploy from Git](#) | [Managed Postgres add-on](#)

Built for Ethiopia's Financial Communities

Transparent

Automated

Scalable

Secure

Thank you

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